Digital Communication System

Multimedia Transmission over Noisy Channels

Project Title:

Design and Implementation of a Unified Digital Communication System for Multimedia Transmission over Wireless Channels

Project Report

Subject:

Communication Systems Lab 2026 (EEP3010)

Group 3

NITESH	(B23EE1047)
NITIN VERMA	(B23EE1048)
NALLAGONDA DHARSHAN	(B23EE1044)
NAGMA SAJ	(B23EE1043)
PALAVALASA VARSHITH NARAYAN	(B23EE1049)

Department of Electrical Engineering

Indian Institute of Technology Jodhpur

2026

Abstract

This project presents the implementation and analysis of a digital communication system designed for the transmission of multimedia signals (image and audio) over noisy channels. Binary Phase Shift Keying (BPSK) was implemented using hardware tools such as LabVIEW and Software Defined Radio (SDR). Higher-order modulation schemes including QPSK and QAM were evaluated through MATLAB-based simulations. System performance was assessed using Bit Error Rate (BER) under Additive White Gaussian Noise (AWGN) conditions. Additionally, the impact of pulse shaping filters, namely Rectangular and Root Raised Cosine (RRC) filters with varying roll-off factors ($\alpha = 0.25, 0.5, 0.75$), was studied.

System Overview

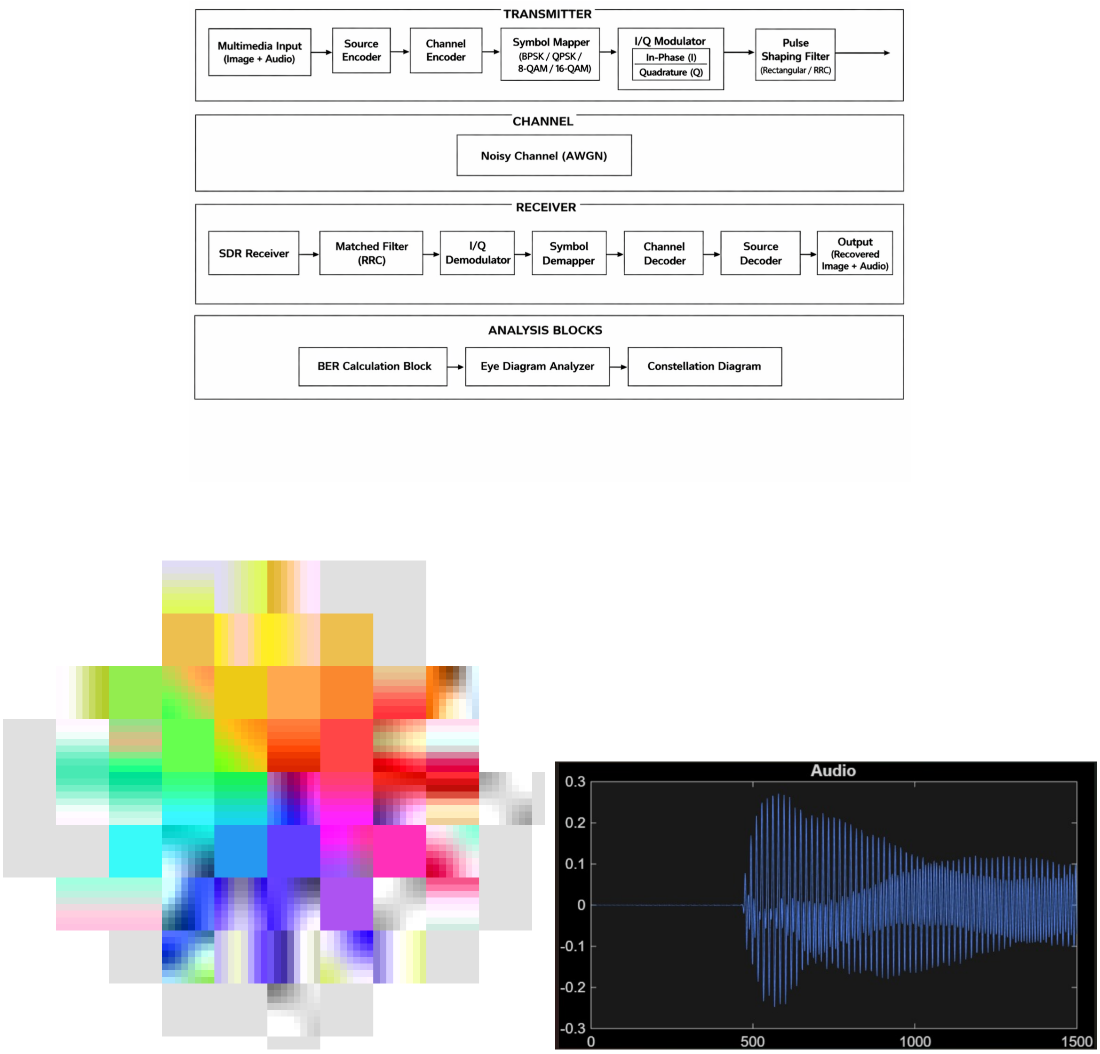

Figure 1: System block diagram along with transmitted image and transmitted audio signal representation.

Theory

BPSK (Binary Phase Shift Keying)

Binary Phase Shift Keying (BPSK) is the simplest digital modulation scheme in which each bit is mapped to one of two phases (0 or π). It is highly robust to noise and commonly used in low-SNR environments.

The transmitted signal can be expressed as:

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi b), \quad b \in \{0, 1\} \tag{1}$$

The Bit Error Rate (BER) for BPSK over an AWGN channel is:

$$BER = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \tag{2}$$

BPSK offers excellent error performance but has low spectral efficiency (1 bit/symbol).

QPSK (Quadrature Phase Shift Keying)

Quadrature Phase Shift Keying (QPSK) improves bandwidth efficiency by transmitting 2 bits per symbol using four different phase states (e.g., 0, $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$).

The signal can be written as:

$$s(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + \theta_i) \tag{3}$$

where $\theta_i \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$.

QPSK can also be viewed as two orthogonal BPSK systems (I and Q components).

The BER for QPSK in AWGN is:

$$BER = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \tag{4}$$

QPSK doubles spectral efficiency compared to BPSK while maintaining similar error performance.

8-QAM (Quadrature Amplitude Modulation)

8-QAM is a modulation scheme that transmits 3 bits per symbol by combining both amplitude and phase variations. It provides higher spectral efficiency than QPSK but is more susceptible to noise.

Unlike square QAM (e.g., 16-QAM), 8-QAM typically uses a non-uniform constellation (such as circular or cross-shaped) to optimize distance between symbols.

The general transmitted signal is:

$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t) \tag{5}$$

where $I(t)$ and $Q(t)$ are the in-phase and quadrature components.

The Bit Error Rate (BER) for 8-QAM does not have a simple closed-form expression like BPSK or QPSK, but it lies between QPSK and 16-QAM:

$$BER_{QPSK} < BER_{\text{8-QAM}} < BER_{\text{16-QAM}} \tag{6}$$

Characteristics:

- 3 bits per symbol
- Higher data rate than QPSK
- Lower noise immunity than QPSK
- More efficient than BPSK and QPSK, but less robust

Trade-off: 8-QAM offers a compromise between spectral efficiency and error performance. It is useful when moderate data rates are required without significantly increasing complexity.

16-QAM (Quadrature Amplitude Modulation)

16-QAM is a higher-order modulation scheme that transmits 4 bits per symbol by varying both amplitude and phase.

The constellation consists of 16 points arranged in a grid pattern. While it improves data rate, it is more sensitive to noise and requires higher SNR.

The approximate BER for 16-QAM in AWGN is:

$$BER \approx \frac{3}{8} Q\left(\sqrt{\frac{4E_b}{5N_0}}\right) \tag{7}$$

Higher-order QAM schemes offer better spectral efficiency but degrade in performance under noisy conditions.

Root Raised Cosine (RRC) Filter

The Root Raised Cosine (RRC) filter is used for pulse shaping in digital communication systems to minimize inter-symbol interference (ISI).

The overall system response (transmitter + receiver) forms a Raised Cosine filter, satisfying the Nyquist criterion for zero ISI.

The frequency response depends on the roll-off factor α:

- $\alpha = 0$: Ideal brick-wall filter (minimum bandwidth, high ISI sensitivity)
- $0 < \alpha < 1$: Practical trade-off between bandwidth and ISI
- $\alpha = 1$: Maximum bandwidth, minimal ISI

The bandwidth of the RRC filter is:

$$B = \frac{1+\alpha}{2T} \tag{8}$$

Effect of Roll-off Factor (α):

- Lower $\alpha \rightarrow$ Higher spectral efficiency, but more ISI sensitivity
- Higher $\alpha \rightarrow$ Better time-domain localization, reduced ISI, but increased bandwidth

Eye Diagram

An eye diagram is a visualization tool used to assess signal quality in digital communication.

- Wide eye opening $\rightarrow$ Low ISI and noise
- Closed eye $\rightarrow$ High ISI and distortion

It helps evaluate timing jitter, noise margin, and synchronization performance.

Constellation Diagram

A constellation diagram represents the modulation symbols in the I-Q plane.

- BPSK $\rightarrow$ 2 points
- QPSK $\rightarrow$ 4 points
- 16-QAM $\rightarrow$ 16 points

Noise causes spreading of points, indicating degradation in signal quality.

Bit Error Rate (BER)

BER is the probability of bit error during transmission and is a key performance metric.

$$BER = \frac{\text{Number of erroneous bits}}{\text{Total transmitted bits}} \tag{9}$$

It depends on:

- Signal-to-noise ratio (SNR)
- Modulation scheme
- Channel conditions

Lower BER indicates better system performance.

Code Implementation

Transmitter Code

Transmitter (Example Code)

```
% ==========================================
% --- TRANSMITTER: Image + Audio over RF ---
% ==========================================
clear; clc;

%% USER CONFIG
mod_scheme = ’16-QAM’;   % ’BPSK’,’QPSK’,’8-QAM’,’16-QAM’
filter_type = ’RRC75’;
samplesPerSymbol = 2;

if strcmp(mod_scheme,’8-QAM’)
    N = 63;
else
    N = 64;
end

%% IMAGE
img = imread(’input_image.jpg’);
if size(img,3)==3
    img = rgb2gray(img);
end
img = imresize(img,[32 32]);
[rows, cols] = size(img);

img_bits = reshape(dec2bin(img(:),8).’-’0’,[],1);

%% AUDIO
[audio, Fs] = audioread(’input_audio.wav’);
audio = audio(:,1);                         % mono
audio = audio(1:min(end,8000));             % limit size
audio_q = uint8((audio+1)*127.5);           % quantize
audio_bits = reshape(dec2bin(audio_q,8).’-’0’,[],1);

audio_len = length(audio_q);
audio_len_bits = reshape(dec2bin(audio_len,16).’-’0’,[],1);

%% HEADER
h_pn = comm.PNSequence(’Polynomial’,[6 5 0],...
    ’InitialConditions’,[0 0 0 0 0 1],...
```

Receiver Code

Receiver (Example Code)

```
% ==========================================
% --- RECEIVER: Image + Audio over RF ---
% ==========================================
clear; clc; close all;

%% CONFIG
mod_scheme = ’16-QAM’;
filter_type = ’RRC75’;
samplesPerSymbol = 2;

if strcmp(mod_scheme,’8-QAM’)
    N = 63;
else
    N = 64;
end

h_pn = comm.PNSequence(’Polynomial’,[6 5 0],...
    ’InitialConditions’,[0 0 0 0 0 1],...
    ’MaximumOutputSize’,[N 1]);

sync_bits = h_pn(N);

%% READ RX
RX = csvread(’Acquired_data.csv’);
RX = complex(RX(:,1), RX(:,2));

%% FILTER
FIR = rcosdesign(0.75,8,samplesPerSymbol,’sqrt’).’;
RX = conv(FIR,RX);

%% DOWNSAMPLE
Symbols = RX(1:samplesPerSymbol:end);
Symbols = Symbols / sqrt(mean(abs(Symbols).^2));

%% DEMOD
switch mod_scheme
    case ’QPSK’, M=4; bps=2;
    case ’8-QAM’, M=8; bps=3;
    case ’16-QAM’, M=16; bps=4;
end
```

Tasks (Modulation + Filter Combinations)

Task 1: BPSK + Rectangular Filter (Hardware)

Image Transmission

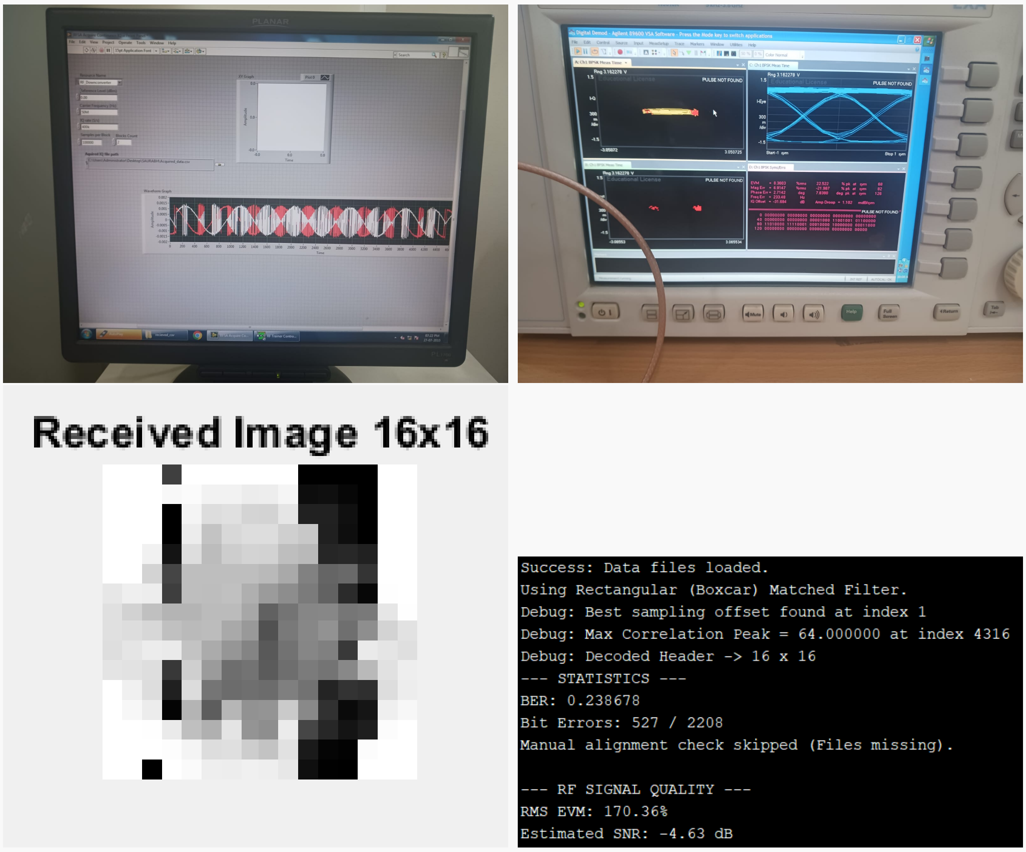

Figure 2: Experimental setup and results for image transmission using BPSK.

Audio Transmission

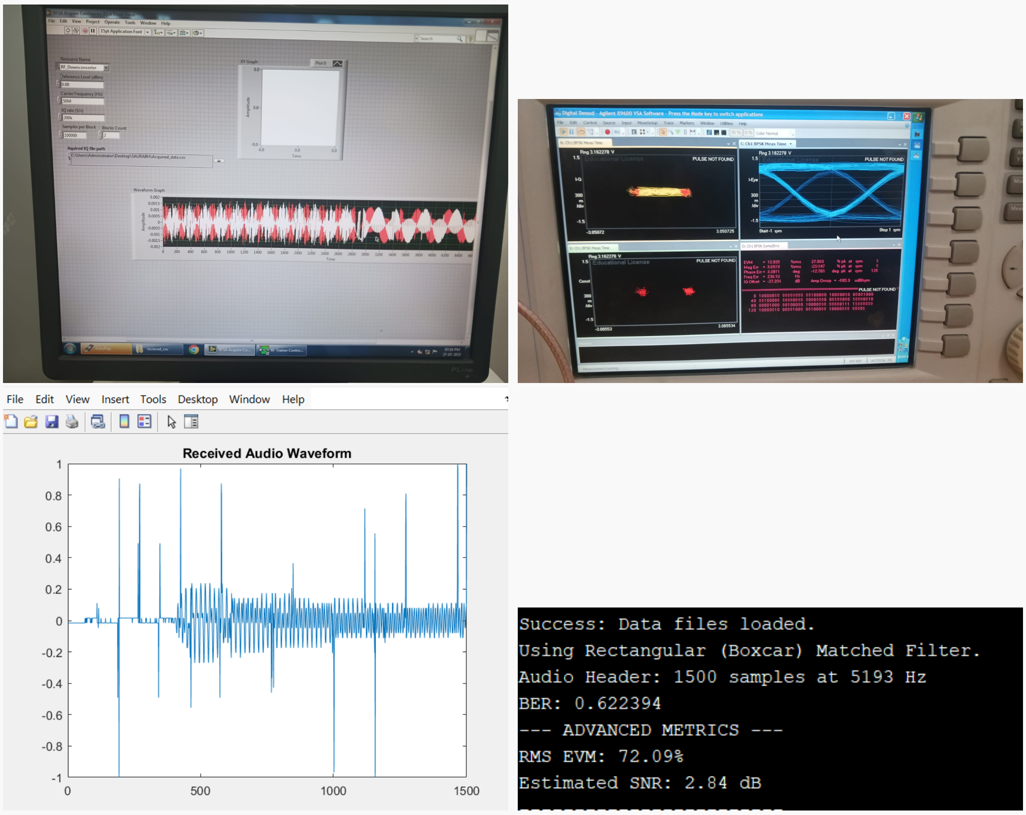

Figure 3: Audio signal transmission and recovery performance.

Task 2: BPSK + RRC Filter $\alpha = 0.25 (Hardware)$

Image Transmission

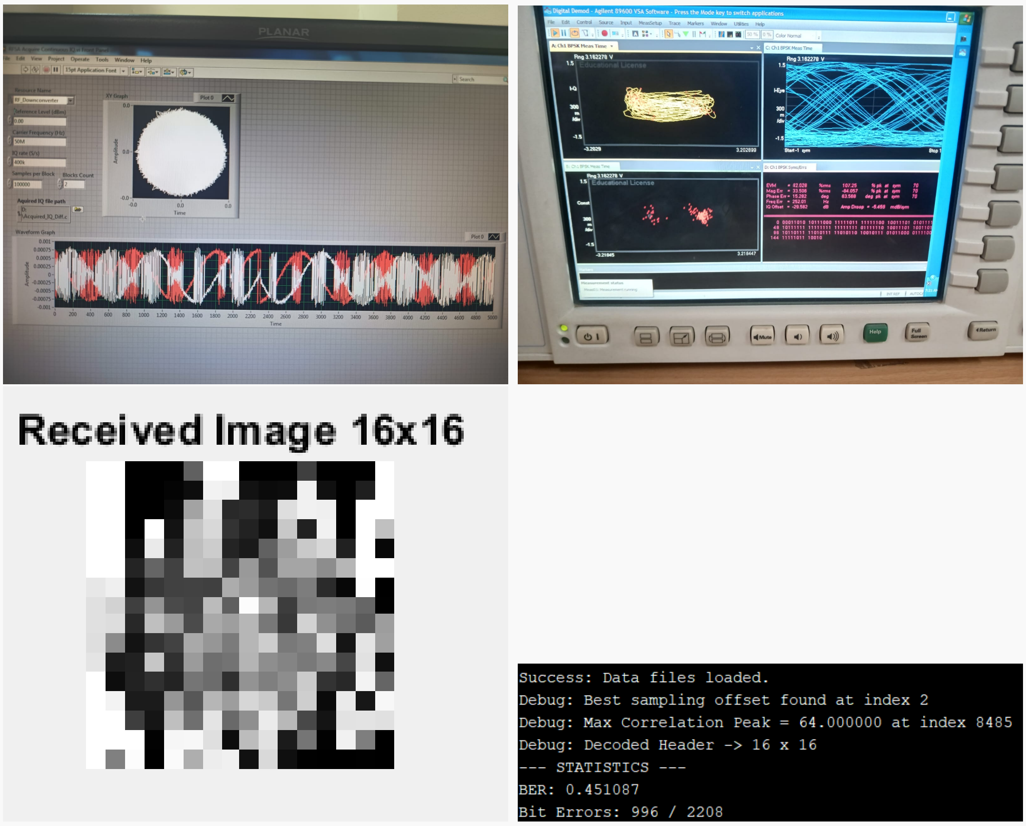

Figure 4: Experimental setup and results for image transmission using BPSK.

Audio Transmission

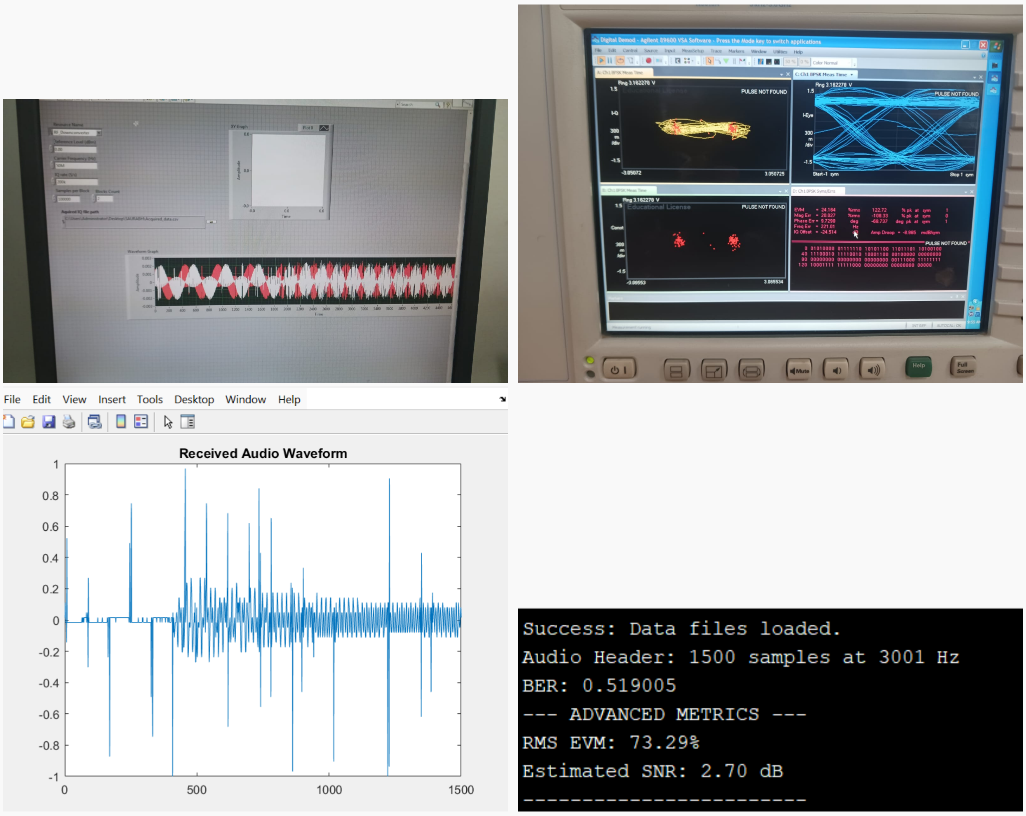

Figure 5: Audio signal transmission and recovery performance.

Task 3: BPSK + RRC Filter $\alpha = 0.50 (Hardware)$

Image Transmission

Figure 6: Experimental setup and results for image transmission using BPSK.

Audio Transmission

Figure 7: Audio signal transmission and recovery performance.

Task 4: BPSK + RRC Filter $\alpha = 0.75(Hardware)$

Image Transmission

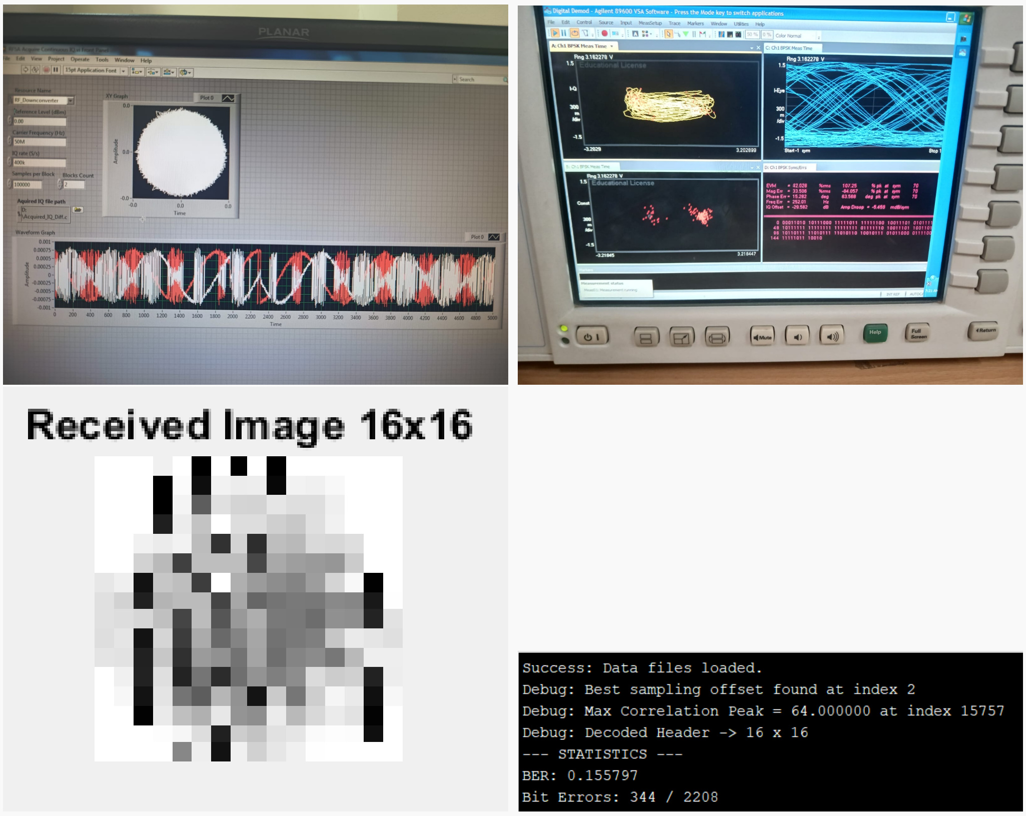

Figure 8: Experimental setup and results for image transmission using BPSK.

Audio Transmission

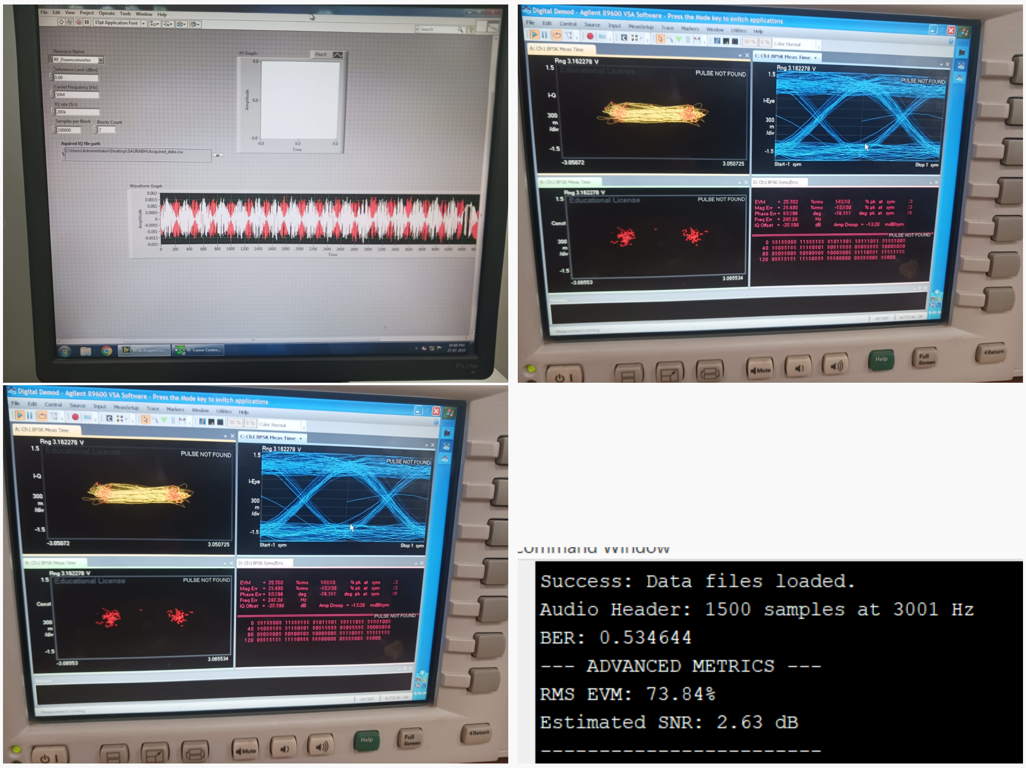

Figure 9: Audio signal transmission and recovery performance.

Task 5: QPSK + RRC ($\alpha = 0.75$) (Simulation)

Image Transmission

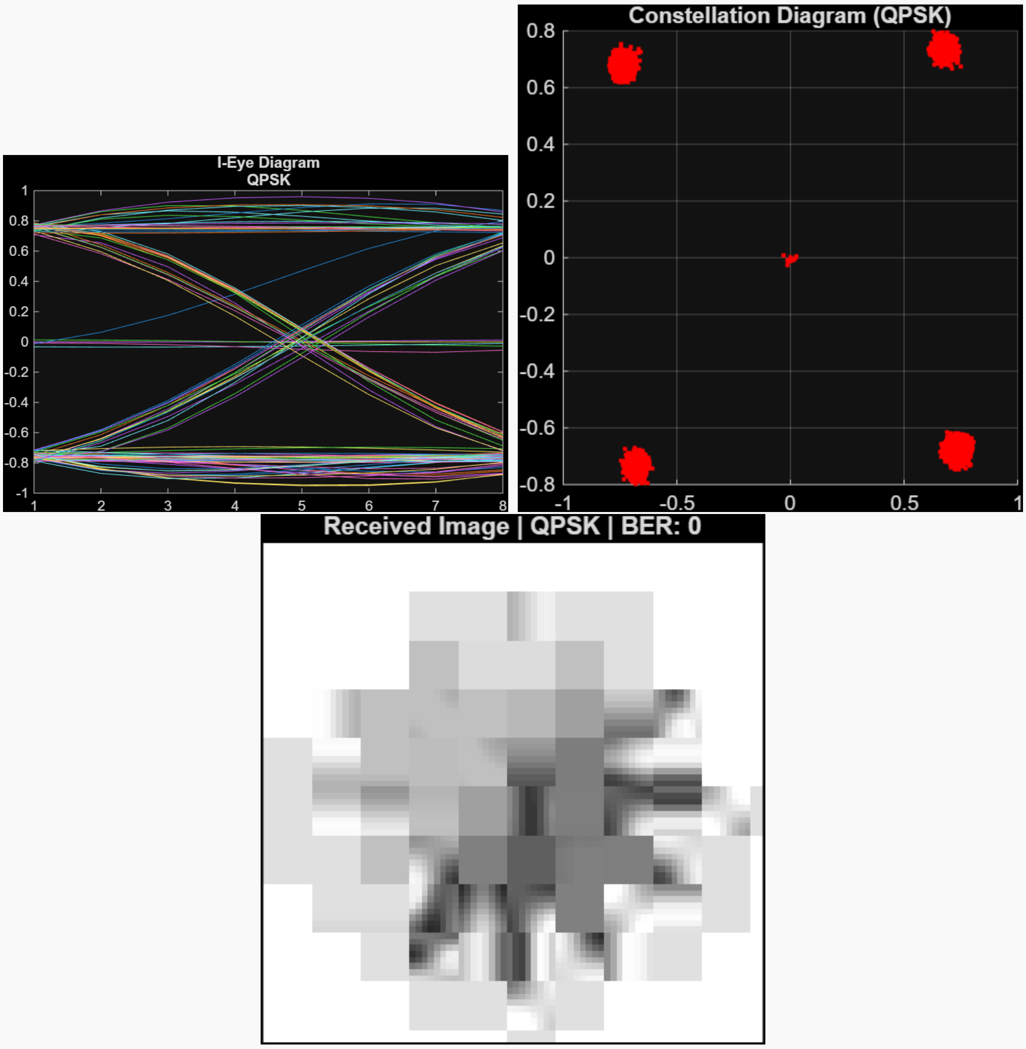

Figure 10: Image transmission results for QPSK with RRC ($\alpha = 0.75$).

Audio Transmission

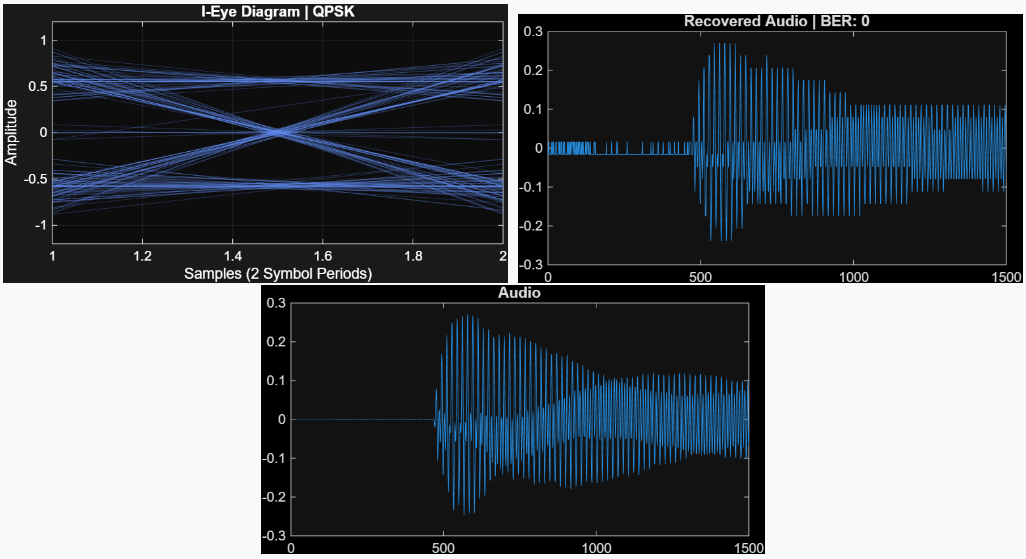

Figure 11: Audio transmission results for QPSK with RRC ($\alpha = 0.75$).

Task 6: 8-QAM + RRC ($\alpha = 0.75$) (Simulation)

Image Transmission

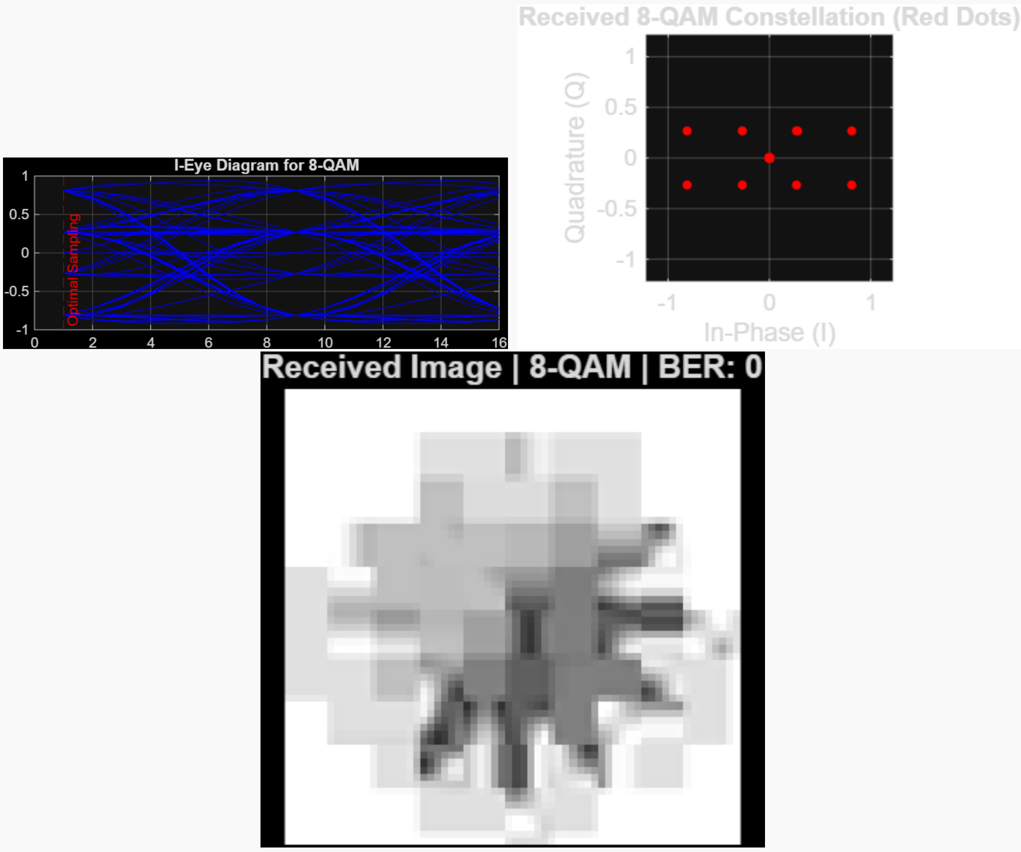

Figure 12: Image transmission results for 8-QAM with RRC ($\alpha = 0.75$).

Audio Transmission

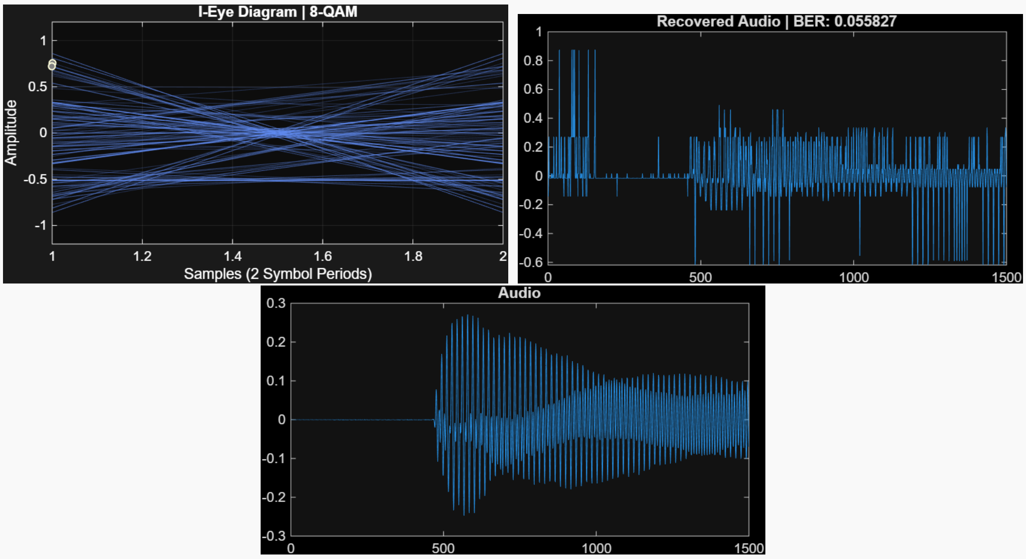

Figure 13: Audio transmission results for 8-QAM with RRC ($\alpha = 0.75$).

Task 7: 16-QAM + RRC ($\alpha = 0.75$) (Simulation)

Image Transmission

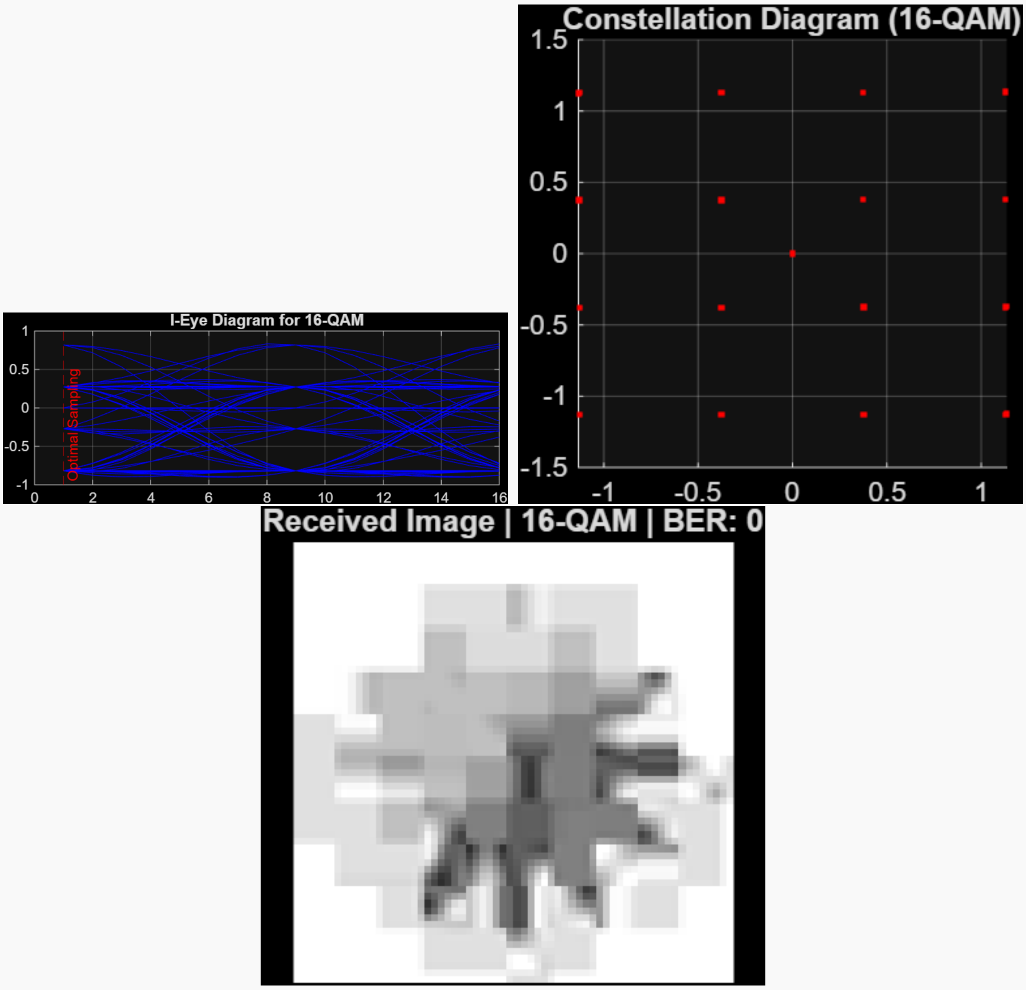

Figure 14: Image transmission results for 16-QAM with RRC ($\alpha = 0.75$).

Audio Transmission

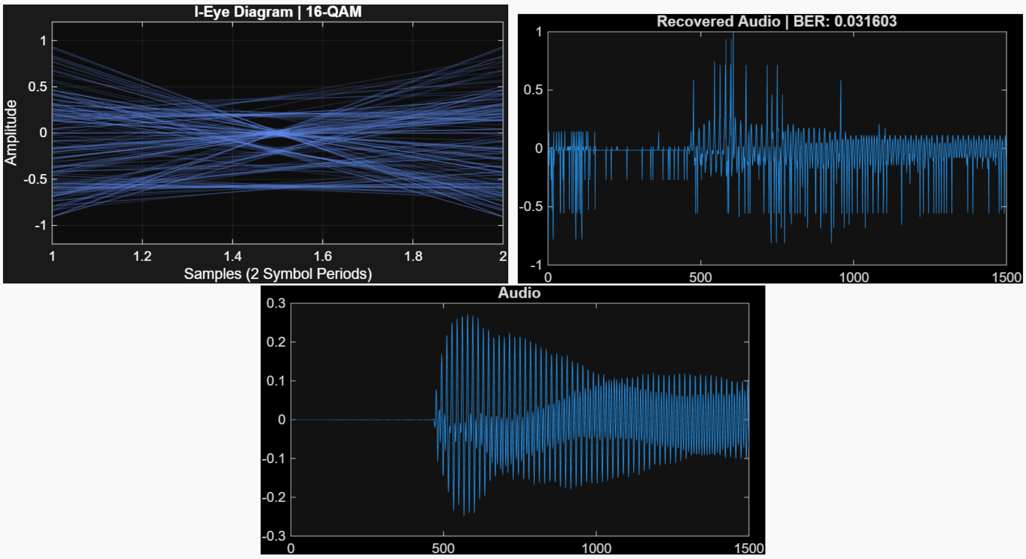

Figure 15: Audio transmission results for 16-QAM with RRC ($\alpha = 0.75$).

BER Comparison

Modulation	Filter	BER
BPSK	Rectangular	0.43
BPSK	RRC (0.25)	0.32
BPSK	RRC (0.5)	0.26
BPSK	RRC (0.75)	0.16

(a) Image Transmission

Modulation	Filter	BER
BPSK	Rectangular	0.53
BPSK	RRC (0.25)	0.50
BPSK	RRC (0.5)	0.45
BPSK	RRC (0.75)	0.43

(b) Audio Transmission

Table 1: BER performance comparison for image and audio transmission.

Discussion

The experimental results are based on the implementation of a BPSK modulation scheme on hardware for both image and audio transmission. From the BER values obtained, it is evident that the use of pulse shaping filters significantly impacts system performance.

For both image and audio transmission, the Rectangular filter shows the highest BER, indicating poor resistance to noise and higher inter-symbol interference (ISI). When Root Raised Cosine (RRC) filters are used, the BER decreases progressively as the roll-off factor increases.

This improvement occurs because higher roll-off factors provide better time-domain localization, reducing ISI and improving symbol detection at the receiver. Among all configurations, the RRC filter with roll-off factor $\alpha = 0.75$ gives the best performance.

Additionally, it is observed that audio transmission results in higher BER compared to image transmission. This is due to the continuous nature and higher sensitivity of audio signals to noise and distortion.

Overall, the results confirm that even with a simple modulation scheme like BPSK, proper filter design plays a crucial role in improving communication reliability.

Challenges

Several challenges were encountered during the hardware implementation:

- **Hardware Constraints:** Only BPSK modulation could be reliably implemented on hardware due to system limitations and stability issues with higher-order modulation schemes.

- **Overheating:** The National Instruments NI PXI downconverter tends to heat up after processing one or two CSV file uploads, requiring a shutdown and restart. This introduces a delay of approximately 5–8 minutes, impacting overall workflow efficiency.

- **Overcrowding:** Multiple groups (around five) were working on similar setups, leading to significant waiting times. On average, each group had to wait nearly an hour to access the equipment, which slowed down overall progress.

- **Synchronization Issues:** Maintaining proper timing synchronization between transmitter and receiver was difficult and led to errors in bit detection.

- **Noise and Interference:** External noise and channel imperfections affected signal quality, especially in audio transmission.

- **Filter Implementation:** Implementing and tuning RRC filters with different roll-off factors required careful parameter selection.

- **Real-Time Processing:** Handling real-time transmission for multimedia data (image and audio) introduced delays and processing challenges.

Conclusion

This project demonstrates the practical implementation of a digital communication system using BPSK modulation for multimedia transmission over a noisy channel. Due to hardware limitations, only BPSK was implemented, but its performance was evaluated under different filtering conditions.

The results show that the use of RRC filters significantly improves system performance by reducing inter-symbol interference and lowering BER. Among the tested configurations, higher roll-off factors provided better reliability.

The project highlights the importance of modulation and filtering techniques in communication systems and provides valuable insights into real-world implementation challenges. Despite limitations, the system successfully achieved reliable transmission of both image and audio data.

Work Distribution

- **NITESH, VARSHITH, NITIN:** Implemented BPSK modulation on hardware, developed transmitter and receiver code, performed BER calculations, and contributed to report writing and final submission.

- **NAGMA, DHARSHAN:** Worked on simulation of higher-order modulation schemes such as 8-QAM and 16-QAM, analyzed their BER performance, and studied the effect of different filter parameters.

- **ALL MEMBERS:** Collaboratively contributed to hardware setup, real-time transmission of image and audio signals, testing, debugging, result verification, and overall system integration.

Project Resources

- **Google Drive (Data & Results):**
  https://drive.google.com/drive/folders/1vA2SJpi1qqcqmbs319GPlzdS8qTmkdKp?usp=sharing

- **GitHub Repository (Code):**
  https://github.com/Nitesh0409/digital-communication-system$_{p}roject$